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Signed By : John Doe
Decision : Approved
Decision Date : 1 Jan 2031 01:00:00 (GMT-04:00)
Role : Approver
Purpose : Review and Approve
Meaning Of Signature : I approve this document as Business Unit.

Signed By : Jane Smith
Decision : Approved
Decision Date : 1 Jan 2031 01:00:00 (GMT-04:00)
Role : Approver
Purpose : Review and Approve
Meaning Of Signature : I approve this document as Business Unit QA/Compliance.

Change History

Release Number	Changes Made -- Reasons for the Change
V1.1	Added names of individuals designated for reviews and approvals

1.0 Introduction

1.1. Purpose

This Test Plan defines the documented testing strategy for **System Testing (ST)** and **User Acceptance Testing (UAT)** to ensure the **Learning Management System (LMS), Version 1.0** is tested with sufficient rigor to demonstrate the system is **fit for intended use** and operates correctly and consistently from a **regulatory and business perspective** throughout the validation lifecycle.
This Test Plan is a **Test Planning deliverable** and is intended to meet the Test Planning requirements defined in **QMS-WI-123, Computer System Validation (CSV) - Testing**, including definition of:

- testing approach and scope,
- test environment (hardware/software), test data, and tools,
- roles and responsibilities,
- training required for test execution,
- traceability expectations, and
- overall acceptance criteria.

This Test Plan also supports the CSV process requirement that validation efforts result in **recorded evidence of activities completed**, and acknowledges that deliverables may be combined based on risk provided required content and approvals are addressed.

System Name:	Learning Management System (LMS)
Configuration Item Number:	CI-LMS-001
Software Version:	1.0 (mock)
Vendor Version (if applicable):	2026.1 (mock)
Testing Plan: (Check as applicable)	☐ System Test (ST) Plan ☐ User Acceptance Test (UAT) Plan ☒ Combined System and User Acceptance Test Plan ☐ Other

1.2. Scope

The scope of testing defined in this Test Plan includes ST and UAT activities required to demonstrate that the LMS meets its **approved requirements** and supports its intended use. Testing scope is driven by the validation strategy and the **Requirements Risk Rating (RRR)** contained within the approved Requirements Specification (RS).

1.2.1. In Scope

The following functional areas and control objectives are **in scope** for LMS Version 1.0 testing:

#	Functions/Enhancements/Bug Fixes	Type of Testing

1	**Core Training Management Business Processes (Intended Use)** • Course and curriculum creation/maintenance (admin functions) • Assignment and scheduling of training to learners • Learner access to assigned training and completion capture • Training completion tracking, status management, and history retrieval	☐ ST ☒ UAT
2	**Electronic Records and Data Integrity Controls** • Creation, modification, and retention of electronic training records • Verification that record outputs (e.g., completion history) are accurate and complete • Controls that prevent unauthorized record modification (role enforcement) • Verification of record protection mechanisms appropriate to system roles	☒ ST ☐ UAT
#	**Functions/Enhancements/Bug Fixes**	**Type of Testing**
3	**Security & Access Controls** • User authentication mechanism for internal users • Role-based access control (admin vs learner vs reporting roles) • Verification of restricted access for privileged functions (configuration/reporting/admin)	☐ ST ☒ UAT
4	**Audit Trail (where implemented for GxP-relevant records)** • Verification that audit trail captures required events for in-scope records • Verification that audit trail entries are attributable and retrievable (as applicable to LMS configuration)	☒ ST ☐ UAT
	Reports & Outputs • **Execution and verification of GxP-relevant training reports (PDF/CSV outputs)** • **Verification of report content accuracy for a defined set of training scenarios**	☒ ST ☐ UAT

1.2.2.Out of Scope

The following items are **out of scope** for this Test Plan and will not be tested under this deliverable:

A. Category 1 infrastructure qualification activities

- Category 1 (Infrastructure) activities are out of scope of the CSV processes defined by the WI/SOP.

B. Disaster Recovery / Business Continuity testing

- Disaster recovery activities are controlled under separate IT procedures and are not included within this Test Plan scope.

C. Performance / Load / Stress testing

- Performance testing is only included if performance requirements are defined; no such requirements are included in this PoC scope, therefore performance/load/stress testing is excluded.

D. Vendor internal regression testing

- Vendor internal development/regression testing is not documented within this LMS Test Plan; any reliance on vendor testing or documentation must be defined in the relevant planning deliverable for hosted solutions.

2.0 Testing Approach

The creation, review, pre-execution approval and post-execution approval of test scripts will be completed in compliance with QMS-WI-2A-2-4, *Computer System Validation (CSV) – Testing*.

Will testing be completed in phases; any overlap between ST and UAT? Specifically, will the UAT begin before ST is completed? ☒ Yes ☐ No **If yes, specify/justify**: System Testing (ST) and User Acceptance Testing (UAT) will be executed in phases, with limited overlap permitted. As a general rule, ST must be completed prior to the start of UAT for the same module or functionality. However, UAT may begin for a specific module once ST has been successfully completed for that module, even if ST activities for other independent modules are still ongoing. This phased approach allows efficient execution while ensuring that no UAT is performed on functionality that has not completed System Testing. All overlaps are controlled, documented, and traceable.
Is there regression testing of functions that are not being changed? ☒ Yes ☐ No **Explain/justify**: Regression testing will be performed on selected LMS functions that are **not directly modified** but may be indirectly impacted by configuration changes or enhancements introduced in this release. Regression testing scope is determined based on risk and includes verification of critical training workflows, user access controls, and reporting functions to ensure no unintended impact to previously validated functionality.
How will the test scripts be executed? ☒ Test scripts will be executed using Application Lifecycle Management (ALM) ☐ Test scripts will be executed on paper ☐ Other

Are changes impacting any business partners who use the system? ☐ Yes ☒ No

If yes, specify/justify: N/A

2.1 System Testing Approach

System Testing is performed to verify that the LMS, as configured and integrated, meets the approved specification set used to drive ST (e.g., Requirements Specification / Functional Requirements) and that key workflows operate as designed under controlled conditions.

System testing will be performed by Team A after the approval of IQ scripts in the test environment. System testing will be performed by Team A qafter the completion of IQ in the test environment.

What are we completing system testing on? Check all that apply:

☒ Core/Whole application

☐ LMS customization

☒ LMS configuration

☐ Other

Why is system testing being conducted on the selected items above?

☐ Custom developed product

☒ Software as a Service

☐ Supplier Assessment results were unsatisfactory

☒ <Regulatory Risk Assessment/SDLC/CSV> identified that system testing is required.

☐ Other

Is the vendor doing ST on behalf of <Client Name>?

☐Yes ☒ No

If yes, then list the following: N/A

What deliverables will be completed to summarize the ST process?

☐ ST Summary Report

☒ Combined ST/UAT Summary Report

Additional Notes:
No separate ST Summary Report will be generated. System Testing results will be documented within the Combined ST/UAT Summary Report. An ST Closure memo is not required as ST completion is formally documented and approved within the combined summary deliverable.

2.2 User Acceptance Testing Approach

User Acceptance Testing is performed to confirm that the LMS meets business/user requirements (intended use) when executed by representative users following defined workflows and acceptance criteria.

Do the groups and locations listed above represent all user areas (departments and sites)? ☒ Yes ☐ No **If not, explain rationale: N/A**
Do the results of the risk assessment or other factors result in requiring additional user acceptance testing? ☐ Yes ☒ No **If yes specify/ justify**: N/A
What deliverables will be completed to summarize the UAT process? ☐ UAT Summary Report ☒ Combined ST/UAT Summary Report
Additional Notes: User Acceptance Testing results will be summarized within the Combined ST/UAT Summary Report. No standalone UAT Summary Report will be generated for this testing effort.

3.0 Test Environment

3.1 Test Environment

Testing will be performed in Dedicated LMS Test Environment which closely reflects the planned production environment.

No separate approved system architecture or design document exists that lists the test environment information; therefore, the test environment details are documented below.
Testing will be performed in the **LMS-TEST** environment. The table below describes the hardware and software requirements for the test environment:

Component	Server Name	Operating System	Software and Version
Web	LMS-WEB-TEST	Linux (RHEL 8.x)	LMS Web UI v1.0 (HTTPS enabled)
Application	LMS-APP-TEST	Linux (RHEL 8.x)	LMS Application Services v1.0
Database	LMS-DB-TEST	Linux (RHEL 8.x)	PostgreSQL 13.x

3.2 Test Data

Test data required for this testing effort will be **synthetic and mock data only**, created specifically to support System Testing (ST) and User Acceptance Testing (UAT).

No production data will be copied or migrated from the production environment into the test environment.

Are a complete or subset of production data being copied from the system's production environment?

☐ Yes ☒ No

Will data be migrated from another system into the testing environment? ☐ Yes ☒ No

If yes, specify/ justify: N/A

Will the data be manually created during/prior to test execution? ☒ Yes ☐ No

Is there another mechanism that is used to load data into the system for this testing effort?

☐ Yes ☒ No

If yes specify/ justify: N/A

Additional Notes:

- Test data will be manually created and configured by the **testing team** before test execution.
- Test data includes synthetic learner accounts, administrator accounts, sample courses, curricula, training assignments, completion records, and report outputs.
- Test data loading is completed **prior to execution of test scripts** and verified as part of test preparation activities.
- No production data is used for this testing effort.
- Test data does **not** contain Personal Identifiable Information (PII).

4.0 Test Tools

The following tools will be utilized to support the testing efforts for **Learning Management System (LMS) Version 1.0**.

Software Tool	CI #	Version	Description of Use
Application Lifecycle Management (ALM)	CI-ALM-001	v2025.4	Used for execution of System Testing and User Acceptance Testing scripts, capture of objective evidence, defect logging, defect tracking, and traceability between requirements, test scripts, and test execution results.
Web Browser (Google Chrome)	N/A	v122.x	Used to access the LMS web application during test execution and to verify user interface functionality and workflows.
Web Browser (Microsoft Edge)	N/A	v122.x	Used as a secondary supported browser to verify LMS compatibility and user workflows during testing.
PDF Viewer	N/A	Latest	Used to review and verify generated LMS reports and exported training records in PDF format.
CSV Viewer / Spreadsheet Tool	N/A	Latest	Used to review and verify LMS report outputs and exported training data in CSV format.

5.0 Test Defect Management

The purpose of this section is to summarize the defect management approach in the event defects are found during the testing phase.

Test defects will be managed in compliance with the applicable CSV Testing procedures. Defects will be documented using a controlled electronic system.

Where will defects be documented and managed?	☒ ALM ☐ Paper ☐ Other. Specify:
Where will defects be tracked if they are not addressed prior to production deployment?	☒ ALM ☐ Service Now ☐ Infinity ☐ Other. Specify.

6.0 Traceability Matrix

How is the traceability matrix being managed?

☐ Project Specific Document

☐ Living Document

☒ Combination Project Specific and Living Document

Additional Notes:

- Traceability will link requirements to test scripts and test execution results using the Application Lifecycle Management (ALM) tool.
- The project-specific traceability matrix will be reviewed and verified upon completion of testing.
- Final traceability confirmation will be documented as part of the Test Summary Report.
- Any requirement or defect not directly traceable to a test script will include documented justification or alternative control.

7.0 Training Plan

The training items listed below are required for members of the project team based on their respective roles and responsibilities in support of testing activities for **Learning Management System (LMS) Version 1.0**.

All personnel involved in System Testing (ST) and User Acceptance Testing (UAT) must be appropriately trained prior to execution of testing activities. Training ensures that testing is performed in a controlled manner and that testing records generated are complete, accurate, and compliant with applicable documentation standards.

Application Lifecycle Management (ALM)

- Training on the use of the Application Lifecycle Management (ALM) tool for:
  - Test script execution
  - Capture of objective evidence (screenshots, logs)
  - Defect logging and tracking
  - Traceability between requirements, test scripts, and execution results
- Review of applicable ALM job aids and quick reference guides prior to test execution

8.0 Test Deliverables

The following are the expected deliverables for this test effort:

Document Name	Planned Storage Location
Test Plan	LMS Validation Document Repository
Approved, Executed test scripts with supporting evidence, where appropriate	Application Lifecycle Management (ALM) tool
Traceability Matrix (required only for validated systems, recommended for non validated systems)	Application Lifecycle Management (ALM) tool
Test Summary Report, including information regarding the resolution of all test defects	LMS Validation Document Repository

9.0 Test Acceptance Criteria

- Test scripts and corresponding objective evidence will be properly documented.
- Testing illustrates that the system meets the intended use as defined in the requirements.
- Test defects will be documented and dispositioned.

10.0 Roles and Responsibilities

The roles and responsibilities for this testing effort are defined in the applicable **Solution Lifecycle Controls (SLC)** procedures (where applicable) and the CSV Testing work instruction. The reviewers and approvers identified below are assigned based on their roles and responsibilities for testing activities associated with **Learning Management System (LMS) Version 1.0**.

Roles may be performed by the same individual where permitted; however, responsibilities related to quality oversight and approval remain independent from test execution activities.

Roles Involved in Testing

- **Test Lead**
  Responsible for planning, coordinating, and managing System Testing (ST) and User Acceptance Testing (UAT) activities. The Test Lead ensures test scripts are prepared, executed, retested as required, and that testing results are accurately documented.

- **Validation Lead**
  Responsible for ensuring that testing activities are executed in accordance with CSV requirements and that testing deliverables align with the overall validation strategy. The Validation Lead reviews and approves testing deliverables as required.

- **Quality Assurance**
  Responsible for reviewing and approving testing deliverables to ensure compliance with regulatory and quality requirements. Quality Assurance provides independent oversight of the testing process.

- **Business Owner**
  Responsible for confirming that the LMS meets intended business use through review and approval of User

Acceptance Testing results. The Business Owner provides acceptance of the system from a business process perspective.

- **Tester**
  Responsible for executing approved test scripts, documenting results, capturing objective evidence, and identifying defects encountered during testing.

11.0 References

Document ID#	Title
QMS-SOP-1	Computer System Validation (CSV)
QMS-SOP-2	GxP Good Documentation Practices

12.0 Reviews and Approvals

For regulated systems, the table below provides guidance on required reviews and approvals for this Test Plan.

Document	Reviewer	Approver
ST Plan	Test Lead	Validation Lead DCM/CI Owner
UAT Plan	Test Lead Validation Lead	Business Owner Quality Assurance
Combined ST/UAT Plan	Test Lead	Validation Lead Quality Assurance DCM/CI Owner Business Owner

For non-regulated systems, refer to Solution Lifecycle Management procedures for reviewers and approvers.

REVIEWS The individual(s) listed below indicate their review for this document.	
Role	**Participant**
Test Lead	John Doe
Validation Lead	Jane Smith

APPROVALS The electronic signatures of the individual(s) listed below indicate their review and approval of this document.	
Role	**Participant**
Validation Lead	Jane Smith
Quality Assurance	Emily Johnson
DCM/CI Owner	Michael Brown
Business Owner	Robert Wilson